SALAR ABBASPOURAZAD

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WORK EXPERIENCE

Apple Inc., Senior Machine Learning Scientist *Sep 2022 - Present*

- Lead pre-training, evaluation and internal distribution of multi-modal foundation models for wearable data (examples 1 & 2 & 3). These models are internally used by >10 research/production teams, have significantly improved and enabled new production features, and have been shipped on Apple devices in Fall 2025
- Owner and lead developer of a general-purpose repository for data processing, distributed training (pre and post), and evaluation (used internally by >10 research/production teams)
  - Dataset sharding, various pre-training frameworks (contrastive learning, joint-embedding architectures, auto-regressive, masked autoencoding, knowledge distillation, etc), various post-training methods (full fine-tuning, LoRA, etc), various architectures and their variants (Transformers, MoEs, Attention, CNNs, etc), multi-node training/inference on Apple secure cloud for health data
  - While being general-purpose, most methods/architectures are tailored for sequential and time-series data , and all are written from scratch in native `pytorch` & `pyspark`
- Worked on multi-modal intelligence via adaptation and instruction tuning of large language models (LLM s) to reason based on time-series data, written with `pytorch` and `transformers`
- Published 5 papers (2 lead author, 3 core author) including ICLR , ICML and Nature Communications . Filed 2 patents (1 lead inventor, 1 co-inventor). Co-mentored 5 interns

Apple Inc., Machine Learning Intern *May 2021-Aug 2021*

Developed contrastive self-supervised deep learning models for signals recorded from wearable devices

EDUCATION

PhD, Electrical & Computer Engineering, University of Southern California, Los Angeles *2015 - 2022*
Thesis: Dynamical representation learning for multiscale brain activity
Developed deep learning and statistical machine learning models for brain machine interfaces
Published 3 papers in Nature journals

MSc, Computer Science, University of Southern California, Los Angeles *2018 - 2020*
GPA: 3.97/4.0
Track: Artificial Intelligence

BSc, Electrical Engineering, Sharif University of Technology, Tehran, Iran *2010 - 2015*
GPA: 18.92/20 (equivalent to 4.0/4.0, class of 2015 top 3 students)
Minor in Economics

HONORS AND ACHIEVEMENTS

Ranked **2nd** among more than 300,000 competitors in the nationwide BSc exam , Iran *2010*

RESEARCH INTERESTS

Deep representation learning, Self-supervised learning, Foundation models pre-training and post-training

SELECT PUBLICATIONS (GOOGLE SCHOLAR)

E. Erturk*, F. Kamran*, **S. Abbaspourazad**, ..., J. Futoma, "Beyond Sensor Data: Foundation Models of Behavioral Data from Wearables Improve Health Predictions", ***ICML***'25 (link)
A. Miller*, J. Futoma*, **S. Abbaspourazad**, ..., G. Sapiro, "A wearable based aging clock associates with disease and behavior", ***Nature Communications***, 2025 (link)

S. Abbaspourazad et al., "Wearable Accelerometer Foundation Models for Health via Knowledge Distillation", *arXiv*, 2024 (link)

S. Abbaspourazad et al., "Large-scale training of foundation models for wearable biosignals", ***ICLR***'24 (link)

S. Abbaspourazad*, E. Erturk*, B. Pesaran and M. Shanechi, "Dynamical flexible inference of nonlinear latent structures in neural population activity", ***Nature Biomedical Engineering***, 2023 (link)(highlighted on the cover, see nature coverage)

S. Abbaspourazad, M. Choudhury, Y. Wong, B. Pesaran and M. Shanechi, "Multiscale low-dimensional motor cortical state dynamics predict naturalistic reach-and-grasp behavior", ***Nature Communications***, 2021 (link)(USC story)

O. G. Sani, **S. Abbaspourazad**, Y. Wong, B. Pesaran and M. Shanechi, "Modeling behaviorally relevant neural dynamics enabled by preferential subspace identification", ***Nature Neuroscience***, 2020 (link)(USC story)

TECHNICAL STRENGTHS

Modeling	Deep Learning, Self Supervised Learning, Time-series and Sequential Data
Tools	Python, PyTorch, Tensorflow, Spark, Airflow, SQL, C++

SIDE PROJECTS

sonarch.ai	Engineered website's agentic RAG system with `LangChain`, `Pinecone` & `Claude Code`

SELECTED GRADUATE COURSES

Representation Learning, Deep Learning, Artificial Intelligence, Analysis of Algorithms, Probabilistic Machine Learning, Estimation Theory, Game Theory, Natural Language Processing, Database Systems

SERVICE

Co-chair of Machine Learning Summit at Apple
Reviewed multiple papers for Nature, ICLR, Neurips, IEEE

INTERESTS

Soccer (former USC Futsal team member), Traveling, Books